

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

April-May, 2013

CL101 Fundamentals of Civil Engineering

Date: 07.05.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Q-3 must be attempted in answer sheet only.

SECTION – I

- Q - 1 (a) Briefly explain water losses. [05]
- (b) (i) Name the branch of civil engineering which deals with design of different structures. [02]
- (ii) Name the branch of civil engineering which deals with construction of various structures and different types of buildings.
- Q - 2 (a) Describe hydrologic cycle with neat sketch and explain different terminologies for the same. [06]
- (b) Enlist the branches of Civil Engineering. Explain any one in detail. [04]
- (c) Give the functions of the following; [04]
- Plinth, Slab, Footing, Column

OR

- Q - 2 (a) What is dam? Explain briefly different types of dams. [06]
- (b) Summarize the duties of a civil engineer. [04]
- (c) Write short note on Aspect and Grouping. [04]
- Q - 3 **Figure-1** shows wall thickness plan. Draw elevation from the face, where arrow is given. Use levels as given with the figure. Assume suitable scale. [14]

OR

- Q - 3 **Figure-1** shows wall thickness plan. Draw section @ AA. Use levels as given with the figure and also provide windows and doors at suitable positions. Assume suitable scale. [14]

SECTION – II

- Q - 4 (a) Differentiate between Prismatic compass and Surveyors compass. [02]
- (b) The length of a survey line was measured with 20 m chain and it worked out to be 530.00 m. It was subsequently found out the chain was 0.04 m too long. What is the [05]

true length of line?

- Q - 5 (a) The perpendicular offsets were taken from a survey line to an irregular boundary line at an interval of 10 m. Calculate area by average ordinate method and trapezoidal rule. [05]

Distance(m)	0	10	20	30	40	50	60	70
Offsets (m)	1.45	1.80	2.20	1.05	3.65	2.75	1.55	2.10

- (b) (1) Define Any Two: Back bearing, Bearing, Magnetic meridian [02]
(2) Convert following quadrantal bearings to whole circle bearings; [03]
(i) S35°E, (ii) S45°W, (iii) N30°W
(c) Prepare a plain scale of RF = 1/500 to show distance of 78.9m. [04]

OR

- Q - 5 (a) How to measure the area of irregular shape? Explain with neat sketch. [05]
(b) The following bearings were observed in running a closed traverse ABCDE. [05]
Calculate the interior angles of the traverse and apply necessary checks.

Line	Fore Bearing
AB	S45°W
BC	S45°E
CD	N45°E
DE	N45°E
EA	N45°W

- (c) Differentiate between plane surveying and geodetic surveying. [04]
Q - 6 (a) (i) Define any two terms; [07]
Bench mark, Back-sight, Line of collimation, Mean sea level
(ii) Discuss the procedure to setting up of a dumpy level at a particular station.
(b) Enlist the road classification based on Nagpur road plan. Explain each of them in detail. [07]

OR

- Q - 6 (a) The following consecutive readings were taken with a dumpy level at every 10m chainage distance; [07]

0.945, 1.015, 2.445, 2.975, 0.475, 0.800, 1.765, 1.830, 0.195, 1.345

The instrument was shifted after fourth and eighth readings. The first reading was taken on Bench mark of 100.000 m RL. Use Height of Instrument method to calculate the reduced levels of all points and apply usual checks.

- (b) (i) Summarize the advantages of railways over road ways. [07]
(ii) Summarize the disadvantages of airways over road ways.

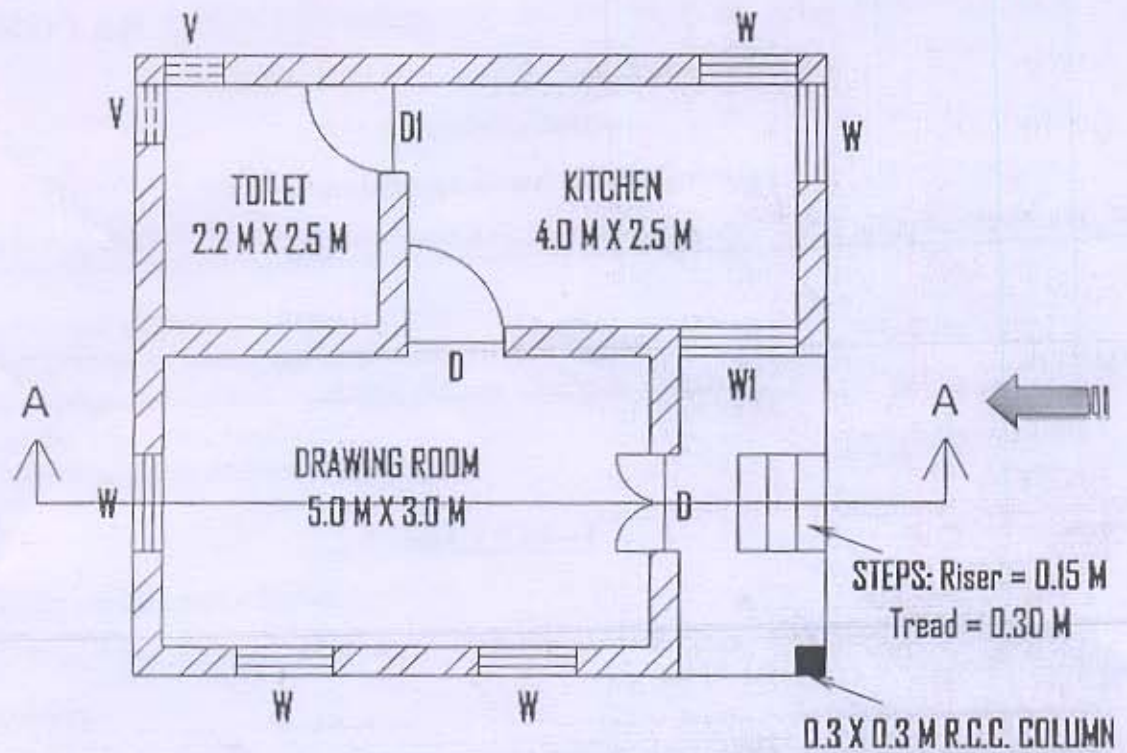


Figure-1

<u>Schedule of Opening:-</u>	<u>Levels :-</u> Plinth level = 0.60 m
Door 'D' = Size : 1.00 x 2.00 = 2 Nos	Lintel level = 2.00 m
Window 'W' = Size : 1.00 x 1.00 = 5 Nos	Floor height = 3.00 m
Window 'W1' = Size : 1.20 x 1.00 = 1 Nos	Parapet height = 0.80 m

Note: All walls having thickness = 0.30 m
All dimensions are in meter.